

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>

Melanie has 5 vacuum cleaners left, which is half of what was left at the orange house before she sold them. Therefore, she had:

$$\backslash[5 \times 2 = 10 \backslash \text{text} \{ \text{vacuum cleaners at the orange house} \} \backslash]$$

Before she went to the orange house, she had sold a third of her vacuum cleaners at the green house and 2 more at the red house, so the number of vacuum cleaners she had left before the orange house was:

$$\backslash[10 + 2 = 12 \backslash \text{text} \{ \text{vacuum cleaners} \} \backslash]$$

Since these 12 vacuum cleaners are two-thirds of the vacuum cleaners she started with (because she had one-third sold at the green house and the rest left), the number of vacuum cleaners she started with is:

$$\backslash[12 \times \frac{3}{2} = 18 \backslash \text{text} \{ \text{vacuum cleaners} \} \backslash]$$

<answer>18</answer>